DARIO LAZZARA

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Education

Università degli Studi di Catania — Current
M.S. in Computer Science: Artificial Intelligence and Machine Learning
- **Focus:** Deep Learning, NLP, Cloud Computing, Social Media Data Analysis.

Università degli Studi di Catania — Sep. 2025
B.S. in Computer Science: Artificial Intelligence & Robotics | **Grade:** 106/110
- **Thesis:** Analyzed **17+ hours of surgical videos** applying **Computer Vision** to extract novel insights, achieving **50–60% recognition accuracy** for hands/instruments.

Università degli Studi di Catania — Feb. 2023
B.S. in Political Science | **Grade:** 108/110
- **Thesis:** Applied **Neural Networks** to 58k survey responses for social data analysis, forecasting migration trends with **85.4%** accuracy.

Technical Skills

Languages & Frameworks: Python, C/C++, SQL, Bash, Pytest

Artificial Intelligence (ML/NLP/CV): PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, GenAI

Data Engineering: Pandas, NumPy, Matplotlib, Playwright, Selenium, Regex

DevOps, Cloud & Tools: AWS, Kubernetes, Docker, Linux, Git, CI/CD, SOLID

Projects

Semantic Alignment of Emojis in Transformer Models — 2026
- Built a semantic alignment pipeline on a **100M tweet corpus**, fine-tuning **RoBERTa** with a Triplet strategy and MNRL to resolve "Context Domination".
- Achieved **90.7% Top-5 Accuracy** in an Information Retrieval task, demonstrating advanced semantic clustering via PyTorch and Hugging Face.

Aspect-Based Sentiment Analysis & LLM Auto-Labeling — 2026
- Engineered an end-to-end scraping and analysis pipeline for banking reviews using **Playwright** and **Gemini 2.0 Flash** for BIO-formatted ground truth labeling.
- Fine-tuned an LCF-ATEPC architecture on an **Italian BERT backbone**, reaching **87.6% sentiment classification accuracy** across multi-aspect reviews.

Image Restoration: Upscaling and Denoising — 2026
- Adapted an **SRGAN architecture** to perform simultaneous **4x upscaling and denoising** on degraded images using adversarial training techniques.
- Fine-tuned the generative network to significantly improve texture recovery and visual clarity, outperforming standard bicubic interpolation baselines.

BoardGuard: Edge AI & IOT Security — 2025
- Deployed an **Edge AI model on ESP32** to detect boat theft via real-time motion and speed sensor analysis with low-latency inference.
- Designed a **two-layer encryption protocol** over a **LoRaMesh** network to secure decentralized long-range vessel communication.

FlyNow (AI Travel Planner) — 2024
- **2nd Place Winner, NeoData Hackathon.** Led a team of 3 to build an AI prototype using **OpenAI APIs**, reducing planning time by 60–70%.

Extracurricular Activity

Quality Development — Nov. 2025
- Developed a Pacman clone in Python applying **Object-Oriented Design (SOLID)** and UNIX shell scripting.
- Implemented unit tests with *pytest* and established **CI/CD pipelines** via GitHub Actions for automated deployment.

Intelligent Agents and Machine Learning — July 2025
- Completed 75 hours of advanced coursework on multi-agent architectures and their applications in AI.
- Developed local multi-agent projects leveraging **LangChain**, **LangGraph**, and **Ollama** for autonomous task execution.

AI & Robotics with Micro-controllers — June 2024
- Mastered core concepts of **TinyML** and robotics for Edge computing, passing the final assessment with a **100% score.**